

1. The binary number $(101.101)_2$ is equal to the decimal number 5.625.
2. A real number has a finite representation in base 2 if and only if it is a rational number.
3. The binary representation of $\frac{1}{3}$ is $(0.\overline{01})_2$.
4. Let $(\bullet)_2$ denote binary representation. Then

Explain briefly:

[illegible]